

GridBalance: Complete Technical Research Dossier

Exhaustive Repository Audit, Data Quality, ML Experiments, Robustness & Production Deployment

UCI Temporal Data Split Report

Chronological Split Protocol

- **Total Valid Samples:** 34,896 hours
- **Training Period:** 2011-01-08 00:00:00 to 2013-12-31 23:00:00 (26,136 samples, ~74.9%)
- **Validation Period:** 2014-01-01 00:00:00 to 2014-06-30 23:00:00 (4,344 samples, ~12.4%)
- **Sealed Test Period:** 2014-07-01 00:00:00 to 2014-12-31 23:00:00 (4,416 samples, ~12.7%)

Note: The test period was kept strictly sealed until final model selection.